

Production planning

Outline

The family

Lot sizing with a fixed set-up cost

Production and workforce: two equivalent formulations

Vehicles: minimum lot and a bonus for variety

The family

Introduction: The family

- ▶ Three problems in which one decides **how much** to produce, not only **whether** to do something
- ▶ The quantity variables are continuous or integer; on top of them sit **binaries** that switch on, limit or reward
- ▶ Problem 1: the production run with its **fixed cost**
- ▶ Problem 2: the **workforce** with the hirings
- ▶ Problem 3: the **minimum lot** per type with a bonus for variety

The common feature: the balance

- ▶ A quantity that comes in, one that goes out and one that stays
- ▶ In problems 1 and 2: the **inventory balance**

$$x_t + s_{t-1} - s_t = d_t$$

- ▶ In problem 2 there is also the **workforce balance**

$$y_t - y_{t-1} - z_t = 0$$

- ▶ These are equality constraints: their duals are **free variables**
- ▶ It is the first family of the course in which this happens systematically

The links between variables that come back here

- ▶ **Fixed cost** (problem 1): $x_t \leq M_t y_t$, with M_t read off the data (the residual demand) and not picked at random
- ▶ **Integer counts** (problem 2): the workforce y_t and the hirings z_t are numbers of people, and the hours available are $r y_t$
- ▶ **Minimum lot** (problem 3): the semi-integer variable $q_j y_j \leq x_j \leq M_j y_j$, which is zero or at least q_j
- ▶ **Counting the types and if and only if** (problem 3): the bonus is collected only if the count of active types reaches two; the missing direction is imposed by optimality

The scheme of every problem

For each of the three problems we shall see:

- ▶ model and proof of the links
- ▶ instance and constructive heuristic (primal bound)
- ▶ dual of the LP relaxation with a hand-built dual solution (dual bound)
- ▶ solver solution and additional considerations
- ▶ a modelling question and a worked variant

All the numbers come from the scripts `fam09_1_lotsizing.py`, `fam09_2_workforce.py` and `fam09_3_vehicles.py`

Notation of the family (1/3): periods and costs

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of periods, $t \in \{1, \dots, n\}$
d_t	$\in \mathbb{Q}_{\geq 0}$	demand of period t
p_t	$\in \mathbb{Q}_{> 0}$	unit production cost
q_t	$\in \mathbb{Q}_{\geq 0}$	fixed set-up cost of a run
h_t	$\in \mathbb{Q}_{\geq 0}$	storage cost at the end of period t
r_0, r_n	$\in \mathbb{Q}_{\geq 0}$	initial and required final inventory
M_t	$\in \mathbb{Q}_{> 0}$	largest useful production: $\sum_{\tau \geq t} d_\tau + r_n$

Notation of the family (2/3): workforce

Symbol	Type	Meaning
m_0	$\in \mathbb{Z}_{\geq 0}$	workers on duty at the start
w	$\in \mathbb{Q}_{>0}$	wage of one worker per period
u	$\in \mathbb{Q}_{\geq 0}$	cost of hiring one worker
r	$\in \mathbb{Q}_{>0}$	hours worked by one worker in one period
g	$\in \mathbb{Q}_{>0}$	labour hours needed for one unit of product

Notation of the family (3/3): types and resources

Symbol	Type	Meaning
s	$\in \mathbb{Z}_{\geq 1}$	number of product types, indexed by j
k	$\in \mathbb{Z}_{\geq 1}$	number of resources, indexed by i
a_{ij}	$\in \mathbb{Q}_{\geq 0}$	amount of resource i per unit of type j
b_i	$\in \mathbb{Q}_{> 0}$	availability of resource i
$\bar{p}_j$	$\in \mathbb{Q}_{> 0}$	unit profit of type j
$\bar{q}_j$	$\in \mathbb{Z}_{\geq 1}$	minimum lot of type j
$\bar{r}$	$\in \mathbb{Q}_{> 0}$	bonus if at least two types are produced

Summary: The family

- ▶ One decides **how much** to produce: quantity variables with binaries that switch on, limit or reward
- ▶ The common feature is the **balance**, written with equality constraints and therefore with free duals
- ▶ Links: fixed cost, integer counts, minimum lot (semi-integer variable), counting the types
- ▶ For every problem: primal bound, dual bound and solver optimum

Lot sizing with a fixed set-up cost

Introduction: Lot sizing with a fixed set-up cost

- ▶ The first problem: a single product to be planned over several periods
- ▶ Every time production is started a **fixed cost** is paid; keeping stock costs
- ▶ We shall see the link $x_t \leq M_t y_t$ and how the **big-M** is read off the data
- ▶ We shall build a primal bound with two heuristics and a dual bound by hand

Problem 9.1: lot sizing with a fixed cost (1/2)

- ▶ A company must plan the production of a single product over $n \in \mathbb{Z}_{\geq 1}$ periods
- ▶ For every period $t \in \{1, \dots, n\}$:
 - ▶ $d_t \in \mathbb{Q}_{\geq 0}$ is the demand to be met
 - ▶ $p_t \in \mathbb{Q}_{> 0}$ the cost of producing one unit
 - ▶ $q_t \in \mathbb{Q}_{\geq 0}$ the fixed cost to be paid if production is started in that period
- ▶ For every $t \in \{1, \dots, n-1\}$, $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one unit in stock from the end of period t to the start of the next one

Problem 9.1: lot sizing with a fixed cost (2/2)

- ▶ The initial inventory is $r_0 \in \mathbb{Q}_{\geq 0}$
- ▶ The one required at the end of the horizon is $r_n \in \mathbb{Q}_{\geq 0}$
- ▶ The company wants to decide how much to produce in every period so as to meet all the demand at **minimum total cost**

The problem in words

- ▶ We **decide** how much to produce in every period, and therefore how much stays in stock
- ▶ **The objective:** minimum total cost (production, set-ups and storage)
- ▶ **The constraints:**
 - ▶ the demand of every period must be met exactly, with what is produced and with what has been kept aside
 - ▶ one cannot produce without paying the fixed set-up cost
- ▶ It is the **lot sizing** problem with set-up

Data (input of the model)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of periods, $t \in \{1, \dots, n\}$
d_t	$\in \mathbb{Q}_{\geq 0}$	demand of period t
p_t	$\in \mathbb{Q}_{> 0}$	unit production cost in period t
q_t	$\in \mathbb{Q}_{\geq 0}$	fixed set-up cost in period t
h_t	$\in \mathbb{Q}_{\geq 0}$	storage cost at the end of period t , $t \leq n - 1$
r_0, r_n	$\in \mathbb{Q}_{\geq 0}$	initial and required final inventory

Decision variables

n non-negative continuous variables:

$$x_t = \text{units produced in period } t, \quad \forall t \in \{1, \dots, n\}$$

$n - 1$ non-negative continuous variables:

$$s_t = \text{units in stock at the end of period } t, \quad \forall t \in \{1, \dots, n - 1\}$$

n binary variables:

$$y_t = \begin{cases} 1 & \text{if production is started in period } t \\ 0 & \text{otherwise} \end{cases}$$

Model 9.1

Model 9.1 — lot sizing with a fixed cost

$$\min \sum_{t=1}^n p_t x_t + \sum_{t=1}^n q_t y_t + \sum_{t=1}^{n-1} h_t s_t \quad (9.1a)$$

$$\text{subject to } x_1 - s_1 = d_1 - r_0, \quad (9.1b)$$

$$x_t + s_{t-1} - s_t = d_t, \quad \forall t \in \{2, 3, \dots, n-1\}, \quad (9.1c)$$

$$x_n + s_{n-1} = d_n + r_n, \quad (9.1d)$$

$$-x_t + M_t y_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.1e)$$

$$x_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.1f)$$

$$s_t \geq 0, \quad \forall t \in \{1, 2, \dots, n-1\}, \quad (9.1g)$$

$$y_t \in \{0, 1\}, \quad \forall t \in \{1, 2, \dots, n\}. \quad (9.1h)$$

Description of the model

- ▶ The **objective function** adds up the production cost, the cost of the set-ups and the storage cost
- ▶ The **inventory balance** constraints are one per period (n equality constraints): what one had plus what one produces minus what one sells is what is left
- ▶ The **set-up** constraints link quantity and set-up (n linear constraints)

The link between the variables (1/2)

The set-up constraint reads

$$x_t \leq M_t y_t$$

- ▶ If $y_t = 0$ then $x_t \leq 0$ and, together with $x_t \geq 0$, one gets $x_t = 0$: **without a set-up nothing is produced**
- ▶ If $y_t = 1$ the constraint becomes $x_t \leq M_t$ and is not restrictive, provided M_t is chosen large enough

The link between the variables (2/2)

- ▶ The other direction, *if $x_t = 0$ then $y_t = 0$* , is not imposed by any constraint
- ▶ It follows from **optimality**: if $x_t = 0$ and $y_t = 1$, setting $y_t = 0$ stays feasible and saves q_t
- ▶ If $q_t > 0$ every optimal solution has $y_t = 0$
- ▶ If $q_t = 0$ the saving is nil: one can only say that there *exists* an optimal solution with $y_t = 0$

The truth table of the link $x_t \leq M_t y_t$

The link must be checked on all the values of y_t , keeping in mind that $x_t \geq 0$:

y_t	constraint	x_t allowed	reading
0	$x_t \leq 0$	only $x_t = 0$	no set-up, no production
1	$x_t \leq M_t$	$0 \leq x_t \leq M_t$	with the set-up one produces

The combination $x_t = 0$ with $y_t = 1$ is left uncovered: it is feasible, but it appears in no optimal solution, because

$y_t = 1$ and $x_t = 0 \implies$ setting $y_t = 0$ stays feasible and saves $q_t \geq 0$

The big-M is read off the data (1/2)

Observation 1: The big-M is read off the data

A valid M_t must be at least the largest quantity that it is worth producing in period t . In an optimal solution one never produces more than the demand still to be covered, final inventory included:

$$M_t = \sum_{\tau=t}^n d_{\tau} + r_n$$

In the first period the initial inventory, which is a datum, can be subtracted: $M_1 = \sum_{\tau=1}^n d_{\tau} + r_n - r_0$.

The big-M is read off the data (2/2)

- ▶ From the second period on the incoming inventory is not known in advance, and the bound stays $\sum_{\tau \geq t} d_{\tau} + r_n$
- ▶ On the instance $r_0 = 0$, so the two values coincide
- ▶ Any larger value is still valid but **weakens the LP relaxation**; any smaller value may **cut off optimal solutions**
- ▶ On the instance $M = (110, 90, 80, 50, 10)$: the big-M of the last period is 10, not 110

The two bounds we shall build, the primal one (heuristic) and the dual one (relaxation), are the two sides of the sandwich that a MIP solver tries to close.

The model in gurobipy (1/2)

```
m = gp.Model("lot_sizing")
x = m.addVars(n, name="x")
s = m.addVars(n - 1, name="s")
y = m.addVars(n, vtype=GRB.BINARY, name="y")
m.setObjective(
    gp.quicksum(p[t] * x[t] for t in range(n))
    + gp.quicksum(q[t] * y[t] for t in range(n))
    + gp.quicksum(h[t] * s[t]
                  for t in range(n - 1)),
    GRB.MINIMIZE)
```

The model in gurobipy (2/2)

```
m.addConstr(x[0] - s[0] == d[0] - r0,
            name="balance[0]")
m.addConstrs((x[t] + s[t - 1] - s[t] == d[t]
              for t in range(1, n - 1)),
             name="balance")
m.addConstr(x[n - 1] + s[n - 2] == d[n - 1] + rn,
            name=f"balance[{n-1}]")
m.addConstrs((-x[t] + M[t] * y[t] >= 0
              for t in range(n)), name="run")
```

The instance

$n = 5$ periods (days), $r_0 = r_n = 0$:

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
d_t	20	10	30	40	10
p_t	2	3	2	3	2
q_t	50	50	50	50	50
M_t	110	90	80	50	10
h_t	1	1	1	1	—

The total demand is 110 units.

Model 9.1 on the instance

Model 9.1 on the instance

$$\begin{aligned}
 &\min 2x_1 + 3x_2 + 2x_3 + 3x_4 + 2x_5 + s_1 + s_2 + s_3 + s_4 + 50y_1 + 50y_2 + 50y_3 + 50y_4 + 50y_5 \\
 &\text{subject to} \quad x_1 \qquad \qquad \qquad -s_1 \qquad \qquad \qquad = 20 \\
 &\qquad \qquad x_2 \qquad \qquad \qquad +s_1 - s_2 \qquad \qquad \qquad = 10 \\
 &\qquad \qquad \qquad x_3 \qquad \qquad \qquad +s_2 - s_3 \qquad \qquad \qquad = 30 \\
 &\qquad \qquad \qquad \qquad x_4 \qquad \qquad \qquad +s_3 - s_4 \qquad \qquad \qquad = 40 \\
 &\qquad \qquad \qquad \qquad \qquad x_5 \qquad \qquad \qquad +s_4 \qquad \qquad \qquad = 10 \\
 &\qquad -x_1 \qquad \qquad \qquad \qquad \qquad \qquad +110y_1 \qquad \qquad \qquad \geq 0 \\
 &\qquad -x_2 \qquad \qquad \qquad \qquad \qquad \qquad \qquad +90y_2 \qquad \qquad \qquad \geq 0 \\
 &\qquad \qquad -x_3 \qquad \qquad \qquad \qquad \qquad \qquad \qquad +80y_3 \qquad \qquad \qquad \geq 0 \\
 &\qquad \qquad \qquad -x_4 \qquad \qquad \qquad \qquad \qquad \qquad \qquad +50y_4 \qquad \qquad \qquad \geq 0 \\
 &\qquad \qquad \qquad \qquad -x_5 \qquad \qquad \qquad \qquad \qquad \qquad \qquad +10y_5 \geq 0 \\
 &\qquad x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5 \qquad \qquad \qquad \geq 0 \\
 &\qquad \qquad \qquad \qquad \qquad \qquad s_1, \quad s_2, \quad s_3, \quad s_4 \qquad \qquad \qquad \geq 0 \\
 &\qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad y_1, \quad y_2, \quad y_3, \quad y_4, \quad y_5 \in \{0, 1\}
 \end{aligned}$$

Dual on the instance

Dual on the instance

$$\begin{array}{llll}
 \max & 20\mu_1 + 10\mu_2 + 30\mu_3 + 40\mu_4 + 10\mu_5 & & \\
 \text{subject to} & \mu_1 & -\pi_1 & \leq 2 \\
 & \mu_2 & -\pi_2 & \leq 3 \\
 & \mu_3 & -\pi_3 & \leq 2 \\
 & \mu_4 & -\pi_4 & \leq 3 \\
 & \mu_5 & -\pi_5 & \leq 2 \\
 & & 110\pi_1 & \leq 50 \\
 & & 90\pi_2 & \leq 50 \\
 & & 80\pi_3 & \leq 50 \\
 & & 50\pi_4 & \leq 50 \\
 & & 10\pi_5 & \leq 50 \\
 & -\mu_1 + \mu_2 & & \leq 1 \\
 & -\mu_2 + \mu_3 & & \leq 1 \\
 & -\mu_3 + \mu_4 & & \leq 1 \\
 & -\mu_4 + \mu_5 & & \leq 1 \\
 & \mu_1, \mu_2, \mu_3, \mu_4, \mu_5 & & \geq 0 \\
 & & \pi_1, \pi_2, \pi_3, \pi_4, \pi_5 & \geq 0
 \end{array}$$

Two things to notice in this dual

- ▶ The constraints on the columns of the stocks,
 $-\mu_t + \mu_{t+1} \leq 1$, say that the price of one unit cannot grow by more than the storage cost from one period to the next
- ▶ The constraint $10\pi_5 \leq 50$ is much looser than the others, because the big-M of the last period is 10 and not 110
- ▶ A small big-M in the primal leaves more room to the dual variable: it is another face of the fact that small big-Ms give **tighter relaxations**

Constructive heuristics: (a) lot-for-lot

Two classic rules, both feasible by construction.

(a) Lot-for-lot: every day one produces exactly the demand of the day

- ▶ no inventory, but a set-up every day

The cost is

$$\sum_{t=1}^5 p_t d_t + \sum_{t=1}^5 q_t = 270 + 250 = 520$$

Constructive heuristics: (b) least unit cost (1/2)

One starts from the first uncovered period and covers with a single run the k periods that minimise the average cost per unit

$$\frac{q + \sum_{i=1}^{k-1} h \cdot (\text{units kept } i \text{ periods})}{\sum_{i=0}^{k-1} d_{t+i}}$$

then one restarts from the first period that is still uncovered.

- ▶ The cost p does not appear: the alternatives start in the same period and cover the same demands, so they pay the same p
- ▶ The average costs are **incremental** costs of set-up plus inventory

Constructive heuristics: (b) least unit cost (2/2)

It is a constructive heuristic: a single solution is built, one element at a time, never backtracking. Running time $O(n^2)$. On the instance:

- ▶ period 1: covers up to period 2, quantity 30, average cost 2
- ▶ period 3: covers up to period 4, quantity 70, average cost ≈ 1.286
- ▶ period 5: covers period 5 alone, quantity 10, average cost 5

Runs are started on days 1, 3, 5; the total cost is 420. Keeping the better of the two heuristics,

$$z(MILP) \leq UB = 420$$

LP relaxation and dual: the dual variables

- ▶ One **free** dual variable μ_t for each balance constraint (equalities)
- ▶ One **non-negative** dual variable π_t for each set-up constraint ($\geq$ direction)
- ▶ Right-hand sides of the balances:

$$b_1 = d_1 - r_0, \quad b_t = d_t \quad (2 \leq t \leq n-1), \quad b_n = d_n + r_n$$

Dual of the LP relaxation of model 9.1

Dual of the LP relaxation of model 9.1

$$\max \sum_{t=1}^n b_t \mu_t \quad (9.2a)$$

$$\text{subject to } \mu_t - \pi_t \leq p_t, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.2b)$$

$$M_t \pi_t \leq q_t, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.2c)$$

$$-\mu_t + \mu_{t+1} \leq h_t, \quad \forall t \in \{1, 2, \dots, n-1\}, \quad (9.2d)$$

$$\mu_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.2e)$$

$$\pi_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}. \quad (9.2f)$$

Description of the dual (1/2)

- ▶ μ_t is the value attached to one unit available in period t ; π_t the price of one unit of set-up capacity
- ▶ The **objective** prices the demand of each period at those values
- ▶ **Columns of the** x_t : producing one unit in period t is worth μ_t and consumes set-up capacity at price π_t ; the balance cannot exceed the unit cost p_t

Description of the dual (2/2)

- ▶ **Columns of the y_t :** all the capacity M_t that a set-up makes available, priced at π_t , cannot be worth more than the fixed cost q_t
- ▶ **Columns of the s_t :** carrying one unit from period t to $t + 1$ earns $\mu_{t+1} - \mu_t$, which cannot exceed h_t
- ▶ The μ_t are free because the balances are equalities; the π_t are non-negative because the set-up constraints are of type $\geq$ in a minimisation

A hand-built dual solution (1/2)

- ▶ Set $\bar{\pi}_t = 0$ for every t : the constraints $M_t \pi_t \leq q_t$ are satisfied because $q_t \geq 0$, and the set-ups become free
- ▶ What is left is $\mu_t \leq p_t$ and $\mu_{t+1} \leq \mu_t + h_t$: the largest feasible value is built forward

$$\bar{\mu}_1 = p_1, \quad \bar{\mu}_t = \min(\bar{\mu}_{t-1} + h_{t-1}, p_t) \quad \text{for } t \geq 2$$

- ▶ $\bar{\mu}_t$ is the lowest unit cost of having one unit available in period t : either by producing it then, or by producing it earlier and keeping it in stock

A hand-built dual solution (2/2)

On the instance $h = 1$ is low and the production costs are 2, 3, 2, 3, 2:

$$\bar{\mu} = (2, 3, 2, 3, 2)$$

$$LB = \sum_{t=1}^5 d_t \bar{\mu}_t = 2 \cdot 20 + 3 \cdot 10 + 2 \cdot 30 + 3 \cdot 40 + 2 \cdot 10 = 270$$

- ▶ It is the production cost if the set-ups were free: a valid bound, and deliberately optimistic
- ▶ The LP relaxation pays for the set-ups in fractions ($\pi_t = q_t/M_t$ is feasible) and goes higher:

$$z(LP) = z(LP^+) = \frac{3890}{11} \approx 353.64$$

Optimal solution

The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

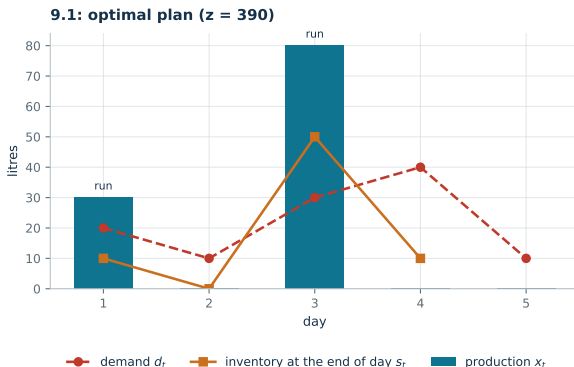
	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
run y_t	1	0	1	0	0
production x_t	30	0	80	0	0
inventory s_t	10	0	50	10	—

The cost is $z(MILP) = 390$, and the chain of bounds is

$$270 = LB \leq z(LP) = \frac{3890}{11} \leq z(MILP) = 390 \leq UB = 420$$

Optimal solution: the gaps and the plan

- ▶ Heuristic gap: $(420 - 390)/390 = 7.7\%$
- ▶ Certified gap, before solving the MILP:
 $(420 - 270)/390 = 38.5\%$



The two runs cover, the first one, days 1–2 and, the second one, days 3–5.

Additional considerations (1/2)

- ▶ The model is a special case of the problem solved in $O(n^2)$ time by the **Wagner–Whitin** dynamic program
- ▶ Least unit cost is *not* that algorithm: it is a myopic rule that looks at one run at a time and in general does not find the optimum (on the instance 420 against 390)
- ▶ The valid inequality $x_t \leq M_t$, obtained from the set-up constraint with $y_t \leq 1$, adds nothing: M_t is already the largest useful production. Indeed $z(LP) = z(LP^+)$

Additional considerations (2/2)

- ▶ An alternative formulation, much tighter, uses the variables

$$x_{t\tau} = \text{units produced in } t \text{ and sold in } \tau$$

(the **facility location** formulation of lot sizing)

- ▶ Its LP relaxation is integral, but it has $O(n^2)$ variables
- ▶ It is the typical trade-off between the **size of the model** and the **quality of the relaxation**

Modelling question: minimum lot

If one produces on a given day, at least 25 units must be produced (below that threshold the set-up is not justified).

How does the model change? What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	420	heuristic solution
LB	270	certified bound
$z(MILP)$	390	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 7.7\%$
- ▶ Certified gap, known *before* solving the MILP (it cannot have $z(MILP)$ as its denominator): $UB - LB = 150$, that is $\frac{UB - LB}{|UB|} = 35.7\%$

A worked variant: daily capacity (1/3)

The plant cannot produce more than 35 units a day. One adds the n linear constraints

$$x_t \leq 35, \quad \forall t \in \{1, \dots, n\}$$

- ▶ Alternatively one replaces the big-M with $M'_t = \min(M_t, 35)$: the constraint then acts both as the activation link and as the capacity constraint, exactly as in capacitated facility location
- ▶ The optimal plan with only two runs is no longer feasible ($80 > 35$): the optimum rises to 470

A worked variant: daily capacity (2/3)

- ▶ The cap $x_t \leq 35$ adds to the dual a family $\nu_t \geq 0$ with right-hand side -35 : it enters the objective with a negative sign and loosens the column of x_t
- ▶ Raising ν_t by one unit lifts the cap on μ_t and collects d_t , but costs 35: it only pays where the demand exceeds the capacity
- ▶ On this instance not even the busiest day justifies it: the certificate stays $LB = 270$
- ▶ What changes is the heuristic: on day four lot-for-lot would exceed the cap, and the 5 excess units are anticipated to day three

A worked variant: daily capacity (3/3)

	value	what it is
UB	520	heuristic solution
LB	270	dual certificate built by hand
$z(LP)$	$\frac{305365}{792}$	relaxation without the bounds
$z(LP^+)$	$\frac{305365}{792}$	relaxation with the bounds
$z(MILP)$	470	optimum of the MILP

Summary: Lot sizing with a fixed set-up cost

- ▶ Inventory balance (equalities) and the link $x_t \leq M_t y_t$; the missing direction is imposed by optimality
- ▶ The big-M is read off the data: $M_t = \sum_{\tau \geq t} d_\tau + r_n$
- ▶ Bounds: $LB = 270$ (free set-ups), $z(LP) \approx 353.64$, $z(MILP) = 390$, $UB = 420$ (least unit cost)
- ▶ The facility location formulation has an integral relaxation but $O(n^2)$ variables

Production and workforce: two equivalent formulations

Introduction: Production and workforce

- ▶ The second problem adds to the production the decision on the **workforce**
- ▶ The staff variables are **integer counts**: numbers of people
- ▶ We shall see two formulations, with the hirings and with the workforce, and prove that they describe the same set of plans
- ▶ It is the first case in the course of two apparently different but equivalent models

Problem 9.2: production and workforce (1/2)

- ▶ A shoe factory must plan its production over $n \in \mathbb{Z}_{\geq 1}$ months
- ▶ For every month t : $d_t \in \mathbb{Z}_{\geq 0}$ is the demand in pairs and $p_t \in \mathbb{Q}_{>0}$ the cost of the raw materials for one pair
- ▶ For every month $t \leq n - 1$: $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one pair in stock at the end of the month
- ▶ At the start the company has $m_0 \in \mathbb{Z}_{\geq 0}$ workers; every worker works r hours a month and costs w euros a month; one pair requires g labour hours

Problem 9.2: production and workforce (2/2)

- ▶ At the start of every month workers may be hired, at a cost of $u \in \mathbb{Q}_{\geq 0}$ euros each
- ▶ Nobody is laid off
- ▶ The company wants to decide how much to produce and how many workers to employ in every month, at **minimum total cost**

The problem in words

- ▶ We **decide** how much to produce, how much to keep in stock and how many workers to hire
- ▶ **The objective:** minimum total cost (raw materials, storage, wages and hirings)
- ▶ **The constraints:**
 - ▶ the demand must be met exactly
 - ▶ the production of a month cannot require more hours than the workers on duty can do

Data (input of the model)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of months, $t \in \{1, \dots, n\}$
d_t	$\in \mathbb{Z}_{\geq 0}$	demand (pairs) of month t
p_t	$\in \mathbb{Q}_{>0}$	cost of the raw materials for one pair
h_t	$\in \mathbb{Q}_{\geq 0}$	storage cost at the end of month t
m_0	$\in \mathbb{Z}_{\geq 0}$	workers on duty at the start
w	$\in \mathbb{Q}_{>0}$	monthly wage of one worker
u	$\in \mathbb{Q}_{\geq 0}$	cost of hiring one worker
r	$\in \mathbb{Q}_{>0}$	hours worked by one worker in one month
g	$\in \mathbb{Q}_{>0}$	labour hours for one pair

Formulation A: the hirings

The staff variables, integer and non-negative, are

$z_t =$ workers hired at the start of month t , $\forall t \in \{1, \dots, n\}$

- ▶ A worker hired in month t stays on duty until the end of the horizon: they cost u once plus w for each of the remaining $n - t + 1$ months
- ▶ The wage of the m_0 initial workers, $m_0 w n$, is a **constant term**: it is left out of the model and added to the final value

Formulation B: the workforce

The staff variables, integer and non-negative, are

$$y_t = \text{workers on duty in month } t, \quad \forall t \in \{1, \dots, n\}$$

- ▶ $y_t \geq y_{t-1}$ because nobody is laid off
- ▶ The cost is $w y_t$ every month, plus u for every hiring, that is $u(y_n - m_0)$ in total
- ▶ The hirings are the increments of the workforce: the telescoping sum leaves only the endpoints

Model 9.2A

Model 9.2A — with the hirings

$$\min \sum_{t=1}^n p_t x_t + \sum_{t=1}^{n-1} h_t s_t + \sum_{t=1}^n (u + w(n - t + 1)) z_t \quad (9.3a)$$

$$\text{subject to } x_1 - s_1 = d_1, \quad (9.3b)$$

$$x_t + s_{t-1} - s_t = d_t, \quad \forall t \in \{2, 3, \dots, n-1\}, \quad (9.3c)$$

$$x_n + s_{n-1} = d_n, \quad (9.3d)$$

$$-g x_t + r \sum_{j=1}^t z_j \geq -r m_0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.3e)$$

$$x_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.3f)$$

$$s_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n-1\}, \quad (9.3g)$$

$$z_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}. \quad (9.3h)$$

Model 9.2B

Model 9.2B — with the workforce

$$\min \sum_{t=1}^n p_t x_t + \sum_{t=1}^{n-1} h_t s_t + \sum_{t=1}^n w y_t + u(y_n - m_0) \quad (9.4a)$$

$$\text{subject to } x_1 - s_1 = d_1, \quad (9.4b)$$

$$x_t + s_{t-1} - s_t = d_t, \quad \forall t \in \{2, 3, \dots, n-1\}, \quad (9.4c)$$

$$x_n + s_{n-1} = d_n, \quad (9.4d)$$

$$-g x_t + r y_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.4e)$$

$$y_1 \geq m_0, \quad (9.4f)$$

$$-y_{t-1} + y_t \geq 0, \quad \forall t \in \{2, 3, \dots, n\}, \quad (9.4g)$$

$$x_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.4h)$$

$$s_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n-1\}, \quad (9.4i)$$

$$y_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}. \quad (9.4j)$$

Description of the two formulations

- ▶ They share the **stock balance** constraints: one per month, what is produced plus what is in stock covers the demand exactly
- ▶ The **available hours** constraints, one per month: the production cannot require more hours than the workers on the payroll can do
- ▶ B has in addition $y_1 \geq m_0$, which starts from the initial workers, and the **monotonicity** constraints, which forbid layoffs
- ▶ In A those same facts are hidden in the domain of the z_t , which are non-negative

The two formulations describe the same problem (1/2)

The correspondence is

$$y_t = m_0 + \sum_{j=1}^t z_j, \quad \text{that is} \quad z_t = y_t - y_{t-1} \quad (\text{with } y_0 = m_0)$$

Feasibility:

- ▶ the hours constraint of A becomes exactly the one of B
- ▶ $z_t \geq 0$ becomes $y_t \geq y_{t-1}$ (monotonicity) and $y_1 \geq m_0$
- ▶ the balances contain no staff variables and stay identical

The correspondence is a **bijection** between the feasible plans of the two models.

The two formulations describe the same problem (2/2)

Cost: the staff cost in B is

$$\sum_{t=1}^n w y_t + u (y_n - m_0) = \sum_{t=1}^n w \left(m_0 + \sum_{j \leq t} z_j \right) + u \sum_{t=1}^n z_t$$

The first term equals $m_0 w n + w \sum_t (n - t + 1) z_t$, because z_j appears in every month from j onwards. Adding the second one:

$$m_0 w n + \sum_{t=1}^n (u + w (n - t + 1)) z_t$$

The two objective functions differ by the constant $m_0 w n$ alone: same optima.

Why keep both

- ▶ A has fewer constraints (no monotonicity) but cost coefficients that depend on the period
- ▶ B has uniform coefficients and extends better if layoffs are added: one only needs $\ell_t \geq 0$ and the balance
$$y_t = y_{t-1} + z_t - \ell_t$$
- ▶ On the instance even the **relaxations** coincide:
$$z(LP) = 15\,960$$
 for both
- ▶ Two formulations that are equivalent on the integers are not always equivalent on the relaxation: the check must be made, not assumed

The model in gurobipy (1/2)

```
m = gp.Model("workforce_B")
x = m.addVars(n, vtype=GRB.INTEGER, name="x")
s = m.addVars(n - 1, vtype=GRB.INTEGER, name="s")
y = m.addVars(n, vtype=GRB.INTEGER, name="y")
m.setObjective(
    gp.quicksum(p[t] * x[t] for t in range(n))
    + gp.quicksum(h[t] * s[t]
                  for t in range(n - 1))
    + gp.quicksum(w * y[t] for t in range(n))
    + u * (y[n - 1] - m0), GRB.MINIMIZE)
```

The model in gurobipy (2/2)

```
m.addConstr(x[0] - s[0] == d[0],
            name="balance[0]")
m.addConstrs((x[t] + s[t - 1] - s[t] == d[t]
              for t in range(1, n - 1)),
             name="balance")
m.addConstr(x[n - 1] + s[n - 2] == d[n - 1],
            name=f"balance[{n-1}]")
m.addConstrs((-g * x[t] + r * y[t] >= 0
              for t in range(n)), name="hours")
m.addConstr(y[0] >= m0, name="initial_workforce")
m.addConstrs((-y[t - 1] + y[t] >= 0
              for t in range(1, n)),
             name="no_layoffs")
```

The instance

$n = 3$ months, $m_0 = 2$ workers, $w = 1500$, $u = 100$, $r = 160$ hours, $g = 4$ hours per pair, $h_t = 3$:

	$t = 1$	$t = 2$	$t = 3$
d_t	60	100	140
p_t	15	15	15

With two workers the initial capacity is $2 \cdot 160/4 = 80$ pairs a month: enough for the first month, not for the other two.

Model 9.2B on the instance

Model 9.2 on the instance

$$\begin{array}{llllllll} \min & 15x_1 + 15x_2 + 15x_3 + 3s_1 + 3s_2 + 1500y_1 + 1500y_2 + 1600y_3 & & & & & & \\ \text{subject to} & x_1 & & -s_1 & & & & = 60 \\ & & x_2 & & +s_1 & -s_2 & & = 100 \\ & & & x_3 & & +s_2 & & = 140 \\ & -4x_1 & & & & & +160y_1 & \geq 0 \\ & & -4x_2 & & & & +160y_2 & \geq 0 \\ & & & -4x_3 & & & +160y_3 & \geq 0 \\ & & & & & y_1 & & \geq 2 \\ & & & & & -y_1 & +y_2 & \geq 0 \\ & & & & & & -y_2 & +y_3 \geq 0 \\ & x_1, & x_2, & x_3 & & & & \in \mathbb{Z}_{\geq 0} \\ & & & & s_1, & s_2 & & \in \mathbb{Z}_{\geq 0} \\ & & & & & & y_1, & y_2, & y_3 \in \mathbb{Z}_{\geq 0} \end{array}$$

Dual of formulation 9.2A on the instance

Dual on the instance

$$\begin{aligned}
 & \max 60\mu_1 + 100\mu_2 + 140\mu_3 - 320\nu_1 - 320\nu_2 - 320\nu_3 \\
 \text{subject to } & \mu_1 & & -4\nu_1 & & \leq 15 \\
 & & \mu_2 & & -4\nu_2 & \leq 15 \\
 & & & \mu_3 & & -4\nu_3 \leq 15 \\
 & -\mu_1 & +\mu_2 & & & \leq 3 \\
 & & -\mu_2 & +\mu_3 & & \leq 3 \\
 & & & & 160\nu_1 + 160\nu_2 + 160\nu_3 & \leq 4600 \\
 & & & & 160\nu_2 + 160\nu_3 & \leq 3100 \\
 & & & & 160\nu_3 & \leq 1600 \\
 & \mu_1, & \mu_2, & \mu_3 & & \geq 0 \\
 & & & & \nu_1, & \nu_2, & \nu_3 \geq 0
 \end{aligned}$$

Constructive heuristic: the primal bound

Just in time production: every month one produces exactly the demand and hires the minimum number of workers needed.

On the instance:

- ▶ month 1: $\lceil 4 \cdot 60/160 \rceil = 2$ workers, no hiring
- ▶ month 2: $\lceil 4 \cdot 100/160 \rceil = 3$, one hiring
- ▶ month 3: $\lceil 4 \cdot 140/160 \rceil = 4$, one more hiring

The cost, including the constant term m_0 with $n = 9000$, is

$$z(MILP) \leq UB = 18\,200$$

Dual of the LP relaxation of model 9.2A

One free variable μ_t per balance, one non-negative ν_t per hours constraint:

Dual of the LP relaxation of model 9.2A

$$\max \sum_{t=1}^n d_t \mu_t - r m_0 \sum_{t=1}^n \nu_t \quad (9.5a)$$

$$\text{subject to } \mu_t - g \nu_t \leq p_t, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.5b)$$

$$-\mu_t + \mu_{t+1} \leq h_t, \quad \forall t \in \{1, 2, \dots, n-1\}, \quad (9.5c)$$

$$r \sum_{t=j}^n \nu_t \leq u + w(n-j+1), \quad \forall j \in \{1, 2, \dots, n\}, \quad (9.5d)$$

$$\mu_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \quad (9.5e)$$

$$\nu_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}. \quad (9.5f)$$

Description of the dual (1/2)

- ▶ μ_t is the value of one pair available in month t ; ν_t the price of one working hour in month t
- ▶ The **objective** prices the demand at those values and subtracts $r m_0 \sum_t \nu_t$: the hours the m_0 initial workers supply for free
- ▶ **Columns of the** x_t : producing one pair is worth μ_t and consumes g hours at price ν_t ; the balance cannot exceed p_t

Description of the dual (2/2)

- ▶ **Columns of the s_t :** holding one pair in stock earns $\mu_{t+1} - \mu_t$, which cannot exceed h_t
- ▶ **Columns of the z_j :** a worker hired at the beginning of month j supplies r hours in each month from j onwards
- ▶ Their total value cannot exceed what that hire costs, namely $u + w(n - j + 1)$

A hand-built dual solution

- ▶ Set $\bar{\nu}_t = 0$: the working hours are not priced, the constraints of the z_j are satisfied; what is left is $\mu_t \leq p_t$ and $\mu_{t+1} \leq \mu_t + h_t$
- ▶ As in problem 9.1: $\bar{\mu}_1 = p_1$, $\bar{\mu}_t = \min(\bar{\mu}_{t-1} + h_{t-1}, p_t)$

On the instance $\bar{\mu} = (15, 15, 15)$ and

$$\sum_{t=1}^n d_t \bar{\mu}_t = 15 \cdot 300 = 4500, \quad LB = 4500 + 9000 = 13\,500$$

- ▶ Raw materials plus the wage of the workers already on duty: the extra labour is given away
- ▶ The LP relaxation pays for it in fractions of a person:
 $z(LP) = z(LP^+) = 15\,960$

Optimal solution

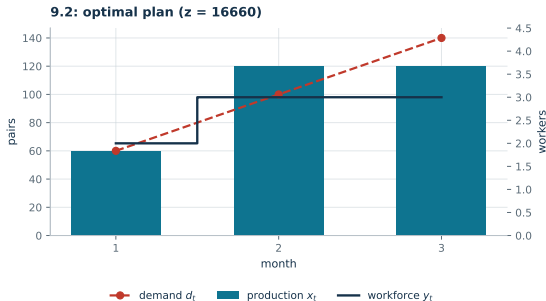
	$t = 1$	$t = 2$	$t = 3$
production x_t	60	120	120
workforce y_t	2	3	3
hirings z_t	0	1	0
inventory s_t	0	20	—

The cost is $z(MILP) = 16\,660$, with the chain

$$13\,500 = LB \leq z(LP) = 15\,960 \leq z(MILP) = 16\,660 \leq UB = 18\,200$$

The heuristic gap is 9.2%.

Optimal solution: the plan



- ▶ The optimal plan hires *one* worker instead of two and moves twenty pairs from the third month to the second
- ▶ 3 euros of storage for twenty pairs cost 60 euros, against the 1600 of a hiring in the third month

Additional considerations (1/2)

- ▶ **Monotonicity** is what ties the months together on the staffing side: if one could lay off at no cost, every month would pick on its own the smallest staff that is enough, and that family would disappear
- ▶ The problem would not split into n independent problems, though: the stocks s_t still tie each month to the next
- ▶ The constant term m_0 w n must be remembered in every comparison: forgetting it makes A look much cheaper than B

Additional considerations (2/2)

- ▶ x_t and s_t are integer because pairs of shoes cannot be split; here they could be left continuous without changing the optimum
- ▶ *Not* because the matrix is totally unimodular: it is not ($z(LP) = 15\,960$ against $16\,660$)
- ▶ The reason is more local: once y is fixed at integer values, what is left in (x, s) is a balance problem with an interval matrix, totally unimodular, and integer right-hand sides (d_t and $r y_t/g = 40 y_t$)
- ▶ It is an argument about the subproblem, not about the whole model

Modelling question: overtime

Every worker may do up to 40 hours of overtime a month,
paid 25 euros an hour.

How does the model change? Is it worth using them?

The final comparison

	value	what it is
UB	18200	heuristic solution
LB	13500	certified bound
$z(MILP)$	16660	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 9.2\%$
- ▶ Certified gap, known *before* solving the MILP:
 $UB - LB = 4700$, that is $\frac{UB - LB}{|UB|} = 25.8\%$

A worked variant: very expensive hirings (1/2)

The hiring cost goes up from 100 to 3000 euros (selection and training).

- ▶ Only the datum u changes: the new optimum is 19 560, that is 2900 euros more
- ▶ The plan does not change: one worker is hired anyway, because without that hiring the problem is infeasible ($4 \cdot 120/160 = 3$ workers)
- ▶ The difference is exactly $3000 - 100 = 2900$: the extra cost of an unavoidable hiring
- ▶ It is an example of sensitivity analysis carried out by re-solving the model

A worked variant: very expensive hirings (2/2)

- ▶ Model, dual and recipe stay those of the base problem; the certificate does not notice the change: the hiring cost does not appear in μ_t , and the bound stays 13500
- ▶ A limiting case: a correct dual bound can be insensitive to the datum that moves the problem; here it is the heuristic that follows it

	value	what it is
UB	24000	heuristic solution
LB	13500	dual certificate built by hand
$z(LP) = z(LP^+)$	17490	relaxation without / with the bounds
$z(MILP)$	19560	optimum of the MILP

Summary: Production and workforce

- ▶ Two equivalent formulations: with the hirings z_t (A) and with the workforce y_t (B), linked by $z_t = y_t - y_{t-1}$
- ▶ The objectives differ by the constant $m_0 w n$: it must be remembered in the comparisons
- ▶ On the instance the relaxations coincide too, but the check must be made
- ▶ Bounds: $LB = 13\,500$, $z(LP) = 15\,960$, $z(MILP) = 16\,660$, $UB = 18\,200$

Vehicles: minimum lot and a bonus for variety

Introduction: Vehicles

- ▶ The third problem is a **maximisation**: profit with limited resources
- ▶ Every type is produced in quantity zero or at least equal to a **minimum lot**: it is the **semi-integer** variable
- ▶ A **bonus** is collected if at least two types are produced: we shall learn to count the active types with one binary and one constraint
- ▶ We shall see that here the relaxation with the bounds beats the hand-built dual

Problem 9.3: vehicles with minimum lot and bonus (1/2)

- ▶ A company produces $s \in \mathbb{Z}_{\geq 1}$ types of vehicle using $k \in \mathbb{Z}_{\geq 1}$ resources
- ▶ $a_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of resource i needed for one unit of type j , and $b_i \in \mathbb{Q}_{>0}$ is the availability of resource i
- ▶ For every type j : $\bar{p}_j \in \mathbb{Q}_{>0}$ is the profit of one unit and $\bar{q}_j \in \mathbb{Z}_{\geq 1}$ the minimum quantity to be produced if that type is chosen

Problem 9.3: vehicles with minimum lot and bonus (2/2)

- ▶ If the production includes at least two different types, the company collects a bonus $\bar{r} \in \mathbb{Q}_{>0}$ (a grant for diversification)
- ▶ The company wants to **maximise the total profit**

The problem in words

- ▶ We **decide** how many units of each type to produce
- ▶ **The objective:** maximum profit, bonus included
- ▶ **The constraints:** the resources must not be exceeded; of a given type one produces either zero units or at least $\bar{q}_j$

Data (input of the model)

Symbol	Type	Meaning
s	$\in \mathbb{Z}_{\geq 1}$	number of types, $j \in \{1, \dots, s\}$
k	$\in \mathbb{Z}_{\geq 1}$	number of resources, $i \in \{1, \dots, k\}$
a_{ij}	$\in \mathbb{Q}_{\geq 0}$	resource i for one unit of type j
b_i	$\in \mathbb{Q}_{> 0}$	availability of resource i
$\bar{p}_j$	$\in \mathbb{Q}_{> 0}$	unit profit of type j
$\bar{q}_j$	$\in \mathbb{Z}_{\geq 1}$	minimum lot of type j
$\bar{r}$	$\in \mathbb{Q}_{> 0}$	variety bonus
M_j	$\in \mathbb{Z}_{\geq 1}$	largest producible amount of type j alone

where $M_j = \min\{\lfloor b_i/a_{ij} \rfloor : a_{ij} > 0\}$.

Decision variables

s non-negative integer variables, s binaries and one binary:

$x_j =$ units produced of type j , $\forall j \in \{1, \dots, s\}$

$$y_j = \begin{cases} 1 & \text{if type } j \text{ is produced} \\ 0 & \text{otherwise} \end{cases} \quad z = \begin{cases} 1 & \text{if the bonus is collected} \\ 0 & \text{otherwise} \end{cases}$$

Model 9.3

Model 9.3 — vehicles with minimum lot and bonus

$$\max \sum_{j=1}^s \bar{p}_j x_j + \bar{r} z \quad (9.6a)$$

$$\text{subject to } \sum_{j=1}^s a_{ij} x_j \leq b_i, \quad \forall i \in \{1, 2, \dots, k\}, \quad (9.6b)$$

$$x_j - \bar{q}_j y_j \geq 0, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.6c)$$

$$x_j - M_j y_j \leq 0, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.6d)$$

$$-\sum_{j=1}^s y_j + 2z \leq 0, \quad (9.6e)$$

$$x_j \in \mathbb{Z}_{\geq 0}, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.6f)$$

$$y_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.6g)$$

$$z \in \{0, 1\}. \quad (9.6h)$$

Description of the model

- ▶ The **objective** adds up the profits of the vehicles produced and the bonus for variety
- ▶ The **resource** constraints, one per resource, are the availabilities
- ▶ The **minimum lot** and **activation** constraints, one per type each, make x_j semi-integer: either zero, or at least $\bar{q}_j$ and at most M_j
- ▶ The **bonus** constraint, a single one, says that the bonus is collected only if at least two types are active

The link between the variables: semicontinuity

The minimum lot and activation constraints together say

$$\bar{q}_j y_j \leq x_j \leq M_j y_j$$

- ▶ If $y_j = 0$ both give $x_j = 0$
- ▶ If $y_j = 1$ one gets $\bar{q}_j \leq x_j \leq M_j$
- ▶ x_j is zero or between the minimum lot and the maximum: it is a **semi-integer** variable

Neither constraint alone is enough: without the activation one a type could be produced with $y_j = 0$; without the minimum lot one the minimum lot would be empty.

The bonus is collected only with at least two types

The bonus constraint reads $2z \leq \sum_{j=1}^s y_j$.

- ▶ If $z = 1$ then $\sum_j y_j \geq 2$: at least two types are activated and, by semicontinuity, actually produced (in quantity $\geq \bar{q}_j > 0$)
- ▶ The other direction is not imposed by the constraint, but follows from **optimality**: if $\sum_j y_j \geq 2$ and $z = 0$, setting $z = 1$ stays feasible and increases the objective by $\bar{r} > 0$
- ▶ Hence in every optimal solution z is a faithful indicator
- ▶ With $\bar{r} = 0$ this argument breaks down, and z stops being faithful

The model in gurobipy (1/2)

```
mm = gp.Model("vehicles")
x = mm.addVars(n, vtype=GRB.INTEGER, name="x")
y = mm.addVars(n, vtype=GRB.BINARY, name="y")
z = mm.addVar(vtype=GRB.BINARY, name="z")
mm.setObjective(
    gp.quicksum(p[j] * x[j] for j in range(n))
    + r * z, GRB.MAXIMIZE)
```

The model in gurobipy (2/2)

```
mm.addConstrs(  
    (gp.quicksum(a[i][j] * x[j] for j in range(n))  
     <= b[i] for i in range(m)), name="resource")  
mm.addConstrs((x[j] - q[j] * y[j] >= 0  
               for j in range(n)), name="minimum_lot")  
mm.addConstrs((x[j] - M[j] * y[j] <= 0  
               for j in range(n)), name="activate")  
mm.addConstr(  
    -gp.quicksum(y[j] for j in range(n)) + 2 * z <= 0,  
    name="bonus")
```

The instance

$s = 3$ types, $k = 2$ resources (steel in tons and labour hours),
 $\bar{q}_j = 10$ for every type, $\bar{r} = 500$:

	$j = 1$	$j = 2$	$j = 3$	b_i
a_{1j} (steel)	2	3	5	100
a_{2j} (hours)	30	25	40	1200
$\bar{p}_j$	200	250	300	
M_j	40	33	20	

The big-Ms are computed from the data:

$M_1 = \min(\lfloor 100/2 \rfloor, \lfloor 1200/30 \rfloor) = 40$, and likewise $M_2 = 33$,
 $M_3 = 20$.

The model for the instance

The instance model

$$\begin{array}{llllllll} \max & 200x_1 & +250x_2 & +300x_3 & & & & +500z \\ \text{subject to} & 2x_1 & +3x_2 & +5x_3 & & & & \leq 100 \\ & 30x_1 & +25x_2 & +40x_3 & & & & \leq 1200 \\ & x_1 & & & -10y_1 & & & \geq 0 \\ & & x_2 & & & -10y_2 & & \geq 0 \\ & & & x_3 & & & -10y_3 & \geq 0 \\ & x_1 & & & -40y_1 & & & \leq 0 \\ & & x_2 & & & -33y_2 & & \leq 0 \\ & & & x_3 & & & -20y_3 & \leq 0 \\ & & & & -y_1 & -y_2 & -y_3 & +2z \leq 0 \\ x_1, & x_2, & x_3 & & & & & \in \mathbb{Z}_{\geq 0} \\ & & & y_1, & y_2, & y_3 & & \in \{0, 1\} \\ & & & & & & z & \in \{0, 1\} \end{array}$$

Constructive heuristic: the primal bound

The problem is a maximisation: the heuristic gives the **primal** bound, which sits below the optimum.

- ▶ The types are sorted by profit relative to the consumption of the tightest resource
- ▶ The two best ones are activated at their minimum lot (so the bonus is guaranteed)
- ▶ Then one fills up with the most profitable among those activated

On the instance types 2 and 1 are activated, then one fills up: the production is (11, 26, 0), the steel runs out and 220 hours are left. The profit is 8700 plus the bonus of 500:

$$z(MILP) \geq LB = 9200$$

LP relaxation and dual: the dual variables

The primal is a maximisation with $\leq$ and $\geq$ constraints; following the conversion table we associate

- ▶ $\pi_i \geq 0$ with the resource constraints
- ▶ $\lambda_j \leq 0$ with the minimum lot constraints
- ▶ $\beta_j \geq 0$ with the activation constraints
- ▶ $\gamma \geq 0$ with the bonus constraint

Writing $\lambda_j = -\ell_j$ with $\ell_j \geq 0$, one gets the dual.

Dual of the LP relaxation of model 9.3

Dual of the LP relaxation of model 9.3

$$\min \sum_{i=1}^k b_i \pi_i \quad (9.7a)$$

$$\text{subject to } \sum_{i=1}^k a_{ij} \pi_i - \ell_j + \beta_j \geq \bar{p}_j, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.7b)$$

$$\bar{q}_j \ell_j - M_j \beta_j - \gamma \geq 0, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.7c)$$

$$2\gamma \geq \bar{r}, \quad (9.7d)$$

$$\pi_i \geq 0, \quad \forall i \in \{1, 2, \dots, k\}, \quad (9.7e)$$

$$\ell_j \geq 0, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.7f)$$

$$\beta_j \geq 0, \quad \forall j \in \{1, 2, \dots, s\}, \quad (9.7g)$$

$$\gamma \geq 0. \quad (9.7h)$$

Dual for the instance

The instance model

$$\begin{array}{llllllll}
 \min & 100\pi_1 + 1200\pi_2 & & & & & & \\
 \text{subject to} & 2\pi_1 & +30\pi_2 & +\alpha_1 & & +\beta_1 & & \geq 200 \\
 & 3\pi_1 & +25\pi_2 & & +\alpha_2 & & +\beta_2 & \geq 250 \\
 & 5\pi_1 & +40\pi_2 & & & +\alpha_3 & & +\beta_3 \geq 300 \\
 & & & -10\alpha_1 & & -40\beta_1 & & -\gamma \geq 0 \\
 & & & & -10\alpha_2 & & -33\beta_2 & -\gamma \geq 0 \\
 & & & & & -10\alpha_3 & & -20\beta_3 -\gamma \geq 0 \\
 & & & & & & & 2\gamma \geq 500 \\
 & \pi_1, & \pi_2 & & & & & \geq 0 \\
 & & & \alpha_1, & \alpha_2, & \alpha_3 & & \leq 0 \\
 & & & & & & \beta_1, & \beta_2, & \beta_3 \geq 0 \\
 & & & & & & & & \gamma \geq 0
 \end{array}$$

Description of the dual (1/2)

- ▶ π_i is the price of one unit of resource i ; ℓ_j and β_j the prices of the two semicontinuity constraints of type j ; γ the price of the bonus
- ▶ The **objective** prices all the available resources at those values
- ▶ **Columns of the** x_j : the resources one unit of type j consumes, priced at π_i and corrected by the two semicontinuity constraints, must cover the profit $\bar{p}_j$

Description of the dual (2/2)

- ▶ **Columns of the y_j :** switching on type j forces at least $\bar{q}_j$ units and allows at most M_j ; the balance of the two prices must cover the bonus γ
- ▶ **Column of z :** the bonus $\bar{r}$ is collected only with two active types, and indeed it has to be covered by 2γ

A hand-built dual solution (1/2)

One proceeds in three steps.

1. The constraint $2\gamma \geq \bar{r}$ forces $\gamma \geq \bar{r}/2$: take the minimum,
 $\bar{\gamma} = \bar{r}/2 = 250$
2. Set $\bar{\beta}_j = 0$ and read from the constraint of the y_j the smallest feasible $\bar{\ell}_j$:

$$\bar{\ell}_j = \frac{\bar{\gamma}}{\bar{q}_j} = \frac{250}{10} = 25 \quad \text{for every } j$$

Every activated type "carries" its share of the bonus

A hand-built dual solution (2/2)

3. What is left is $\sum_i a_{ij} \pi_i \geq \bar{p}_j + \bar{\ell}_j$. One prices *a single* resource, so that it covers all the types, and keeps the one giving the smaller bound:

$$\bar{\pi}_i = \max_j \frac{\bar{p}_j + \bar{\ell}_j}{a_{ij}}, \quad \text{bound} = b_i \bar{\pi}_i$$

On the instance:

$$\bar{\pi}_1 = \max\left(\frac{225}{2}, \frac{275}{3}, \frac{325}{5}\right) = \frac{225}{2}, \quad b_1 \bar{\pi}_1 = 11\,250$$

$$\bar{\pi}_2 = \max\left(\frac{225}{30}, \frac{275}{25}, \frac{325}{40}\right) = 11, \quad b_2 \bar{\pi}_2 = 13\,200$$

The better bound is the one from the steel:
 $z(MILP) \leq UB = 11\,250$.

Optimal solution

- ▶ The optimal production is (26, 16, 0): types 1 and 2 are activated and the bonus is collected
- ▶ All 100 tons of steel are consumed and 1180 hours out of 1200

$$z(MILP) = 200 \cdot 26 + 250 \cdot 16 + 500 = 9700$$

The chain of bounds is

$$\begin{aligned} 9200 = LB \leq z(MILP) = 9700 \leq z(LP^+) = 9750 \\ \leq z(LP) = \frac{20625}{2} \leq UB = 11\,250 \end{aligned}$$

The heuristic gap is 5.2%.

Here the relaxation with the bounds beats the hand-built dual

- ▶ It is the only problem in the chapter in which $z(LP^+) = 9750$ is **better** than the hand-built dual bound (11 250), and only slightly worse than the optimum (9700)
- ▶ The relaxation without the bounds lets y_j and z grow above 1, and with them the bonus: $z(LP) = 20625/2 \approx 10\,312$
- ▶ Adding $y_j \leq 1$ and $z \leq 1$ removes exactly that freedom
- ▶ When a model contains rewarded indicators, the relaxation with the bounds is not a detail

Optimal solution: the plan



Units produced per type, with the minimum lot $\bar{q}_j$ marked in red.
Type 3 stays at zero.

Additional considerations

- ▶ The big-M M_j is the largest producible amount of type j *alone*, not of the whole plan: already much tighter than an arbitrary constant ($M_3 = 20$: the steel alone would allow 20 units, the hours 30)
- ▶ If $\bar{q}_j > M_j$, type j is impossible to produce and is removed by a check on the data: a pre-processing step always worth doing
- ▶ The bonus is modelled with *one* binary and *one* constraint; with a threshold of "at least f types" one replaces 2 with f ; with bonuses in brackets one needs one variable per bracket and a chain of constraints

Modelling question: zero bonus

The diversification grant is abolished, that is $\bar{r} = 0$.

What happens to the variable z ?

The final comparison

	value	what it is
LB	9200	heuristic solution
UB	11250	certified bound
$z(MILP)$	9700	MILP optimum

- ▶ Gap of the heuristic: $\frac{|LB - z(MILP)|}{|z(MILP)|} = 5.2\%$
- ▶ Certified gap, known *before* solving the MILP:
 $UB - LB = 2050$, that is $\frac{UB - LB}{|LB|} = 22.3\%$

A worked variant: a more demanding bonus (1/2)

The bonus is collected only if at least **three** different types are produced. Only one coefficient changes:

$$-\sum_{j=1}^s y_j + 3z \leq 0$$

- ▶ The new optimum is 9200, that is 500 less: activating the third type would cost more than the bonus, and at the optimum $z = 0$
- ▶ The value coincides, by coincidence, with that of the heuristic solution of the original problem

A worked variant: a more demanding bonus (2/2)

- ▶ In the dual only the column of z changes: $\gamma = \bar{r}/3 = 500/3$, and the bound drops from 11250 to $32500/3$
- ▶ Three types instead of two spread the bonus over more activations: each type carries less of it
- ▶ The heuristic does not manage to switch on a third type: it stays at 8700

	value	what it is
UB	$\frac{32500}{3}$	dual certificate built by hand
LB	8700	heuristic solution
$z(LP) / z(LP^+)$	$\frac{29875}{3} / \frac{28750}{3}$	relaxation without / with the bounds
$z(MILP)$	9200	optimum of the MILP

Summary: Vehicles

- ▶ **Semi-integer** variable: $\bar{q}_j y_j \leq x_j \leq M_j y_j$, with M_j computed from the data
- ▶ Counting the types: $2z \leq \sum_j y_j$; the missing direction is imposed by optimality because $\bar{r} > 0$
- ▶ The problem is a maximisation: the heuristic gives $LB = 9200$, the hand-built dual $UB = 11\,250$
- ▶ The relaxation with the bounds ($z(LP^+) = 9750$) beats the hand-built dual: with rewarded indicators the bounds $y_j \leq 1$, $z \leq 1$ matter

Final summary (1/2)

Modelling production:

- ▶ quantity variables with binaries that switch on (fixed cost), limit (minimum lot) or reward (variety)
- ▶ the inventory and workforce balances are written with equalities, whose duals are free

Big-M:

- ▶ it is read off the data (residual demand, largest producible amount): larger values weaken the relaxation

Final summary (2/2)

Formulations:

- ▶ two different models can describe the same plans (hirings against workforce); the equivalence on the relaxation must be checked

Bounds:

- ▶ constructive heuristics for the primal bound, hand-built dual solutions for the dual one
- ▶ a dual bound can be insensitive to the datum that moves the problem
- ▶ with rewarded indicators the relaxation with the bounds can beat the hand-built dual